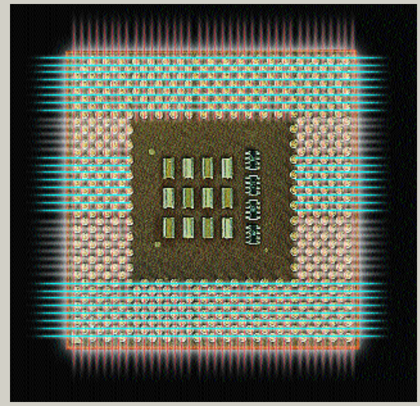
The Optical Processor

The year 2003 was a landmark year in the journey towards optical computing: In March, Lightbit Corp, USA, came up with an optical processor that deals with multiple channels in fibre-optic networks. Lenslet, an Israeli startup, also launched one of its supercomputer-calibre chips the same year. Lenslet said its processor, called Enlight, would enable "new capabilities in homeland security and military, multimedia and communications applications."

Enlight can process multiple DWDM channels in fibre-optic networks simultaneously, without the intervening electronic conversion. Just one of these processors can replace multiple optical-electronic-optical (OEO) transponders. The chip harnesses the 'parallel-processing power of light'.

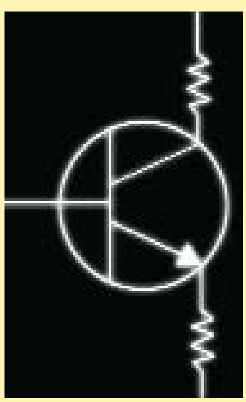
Lenslet's offering is more interesting: It is a digital signal processor (DSP) to which is attached an 'optical accelerator', which enhances the chip's speed. Aviram Sariel, founder and CEO, Lenslet, said this chip was "an acceleration of 20 years in the development of digital hardware," speaking to Reuters in Israel. There are 256 lasers—the active computing elements—within the chip, leading to a performance of eight trillion operations per second. Roughly speaking, this is equivalent to that of a 'regular' supercomputer, and a thousand times faster than a typical commercial processor.

So, has 'the optical processor' arrived? Not quite. These are not general-purpose processors—you cannot use them for anything other than their intended use. So, why is there interest in them? Well, these processors signify a major leap into fully-optical processors, and we are now only a small step away from a general-purpose fully-optical processor, as far as theory is concerned.



Optical Transistors and Logic Gates

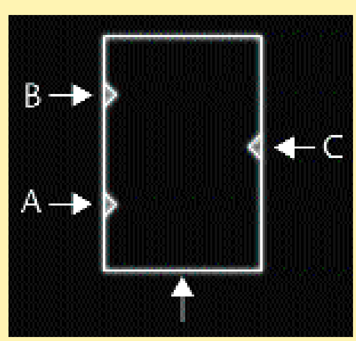
In a way, the logic gate is the basic building block of a computer and transistors are building blocks of logic gates—at least in today's silicon-computer world. The discovery of optical devices that act as transistors is being actively pursued, since it could lead to new possibilities in computer designs. Processors and other chips could be built using designs that were hitherto impossible, on the basis of optical transistors.



So, what's come out of the research labs along these lines? Plenty! A transistor is essentially a switch, and the defining speed parameter here is its switching time. Evident Technologies, USA, has come up with a 'planar lightwave circuit device' that acts as a transistor with a switching time of 1 picosecond. (Regular transistors, such as those in a Pentium chip, have switching times in the order of a few picoseconds.) The switching process takes place using light pulses.

In November 2003, Stanford and MIT researchers brought out designs for an optical switch and an optical transistor. These are less than a micron in size, and their designs can be fabricated in regular chip-fabricating facilities. The devices would work on the principle of the bending, or refracting, of light in a photonic crystal. Researchers said that all-optical devices on chips can be feasible as early as 2005-2008.

Back in March 2003, researchers at the Georgia Institute of Technology developed nanoscale opto-electronic devices, and demonstrated their functioning as logic gates. The principle of these opto-electronic transistors, or switches, is simple: Instead of measuring current to determine device output, one measures the light—or electroluminescent—output. Again here, the researchers—Robert Dickson and Tae-Hee Lee—emphasise that their devices will not replace traditional silicon transistors, but could be used in dedicated devices—ones that do a specific task—such as optical network switches.



This wasn't the first optical transistor: In May 2000, Professors Sajeev John and Geoffrey Ozin of the University of Toronto developed one using silicon photonic crystal. It trapped light waves and controlled their path, much like a silicon transistor manipulating the path of electrons through it.

John and Ozin's transistor highlighted one of the biggest hurdles along the path to the 'photonisation' of computers: The search for materials. There's a lot of theory out there, but materials that actually support the theory are hard to come by. And in the case of John and Ozin, a major part of their efforts focused on producing the silicon photonic crystal.